

2310P: Global Contribution and Representation on Pivotal Clinical Trials in Hematology/Oncology

Anya Pant¹, Fen Saj², Aditya Desai³, Ryan Huey², Jonathan Loree⁴, Kanwal Raghav²

¹Episcopal High School, Houston, TX, USA; ²Gastrointestinal Medical Oncology, The University of Texas MD Anderson Cancer Center, Houston, TX, USA; ³Lakeside School, Seattle, WA, USA; ⁴Medical Oncology, BC Cancer Center, Vancouver, Canada

Plain Language Summary

➤ Why did we perform this research?

- Clinical trials for patients with cancer are performed worldwide. While these trials increase accessibility to clinical research for patients globally, it is not clear how this has affected participation of individual countries/regions.

- We performed this study by looking at participation of patients across the globe in pivotal trials which led to the FDA approval of cancer drugs.

➤ How did we study this research questions?

- We looked at the published reports of pivotal trials in cancer patients for new drugs approved between 6/2022-6/2025.
- We then looked at participation of patients in these trials and compared this to the number of sites by each country/region.

➤ What were the findings of our research study?

- Nearly half of all trials did not report on the number of patients by their country/region.
- In our study, the relative contribution of high-income countries like US/Canada was lower than expected, based on the proportion of trial sites.

➤ What is the key takeaway from this research?

- With globalization of clinical trials, it is important to ensure that patients across the world are well represented so that results of trials are generalizable to the population.

References

- Barrios CH, et al. Am Soc Clin Oncol Educ Book:e132-9, 2015
- Moyers JT et al. Cancer Discov 12:886-888, 2022
- Loree JM et al. JAMA Oncol 5:e191870, 2019
- <https://www.cancernetwork.com/view/fda-odac-votes-8-to-1-against-applicability-of-glofitamab-dlbc-data>

Introduction

- Hematology and Oncology trials have undergone rapid globalization, with a shift from academic centers in high-income countries (HICs) to hundreds of sites across diverse income regions, worldwide.¹
- While globalization potentially accelerates enrollment, increases accessibility and broadens market access, it can paradoxically alter relative participation among different regions.²
- Underrepresentation of populations in clinical trials, based on race and regional imbalances can mar the generalizability of findings.³
- Recently, the Food and Drug Administration (FDA) has emphasized US representation in clinical trials.
- Drug applications have had unfavorable outcome, failing to meet this standard.⁴
- We therefore investigated as to how globalization has altered geographic representation in pivotal hematology/oncology trials, hypothesizing that North America's patient contribution is disproportionately less relative to its site representation.

Methods

- We conducted a systematic review of all Hematology and Oncology FDA drug approvals between June 22, 2022, and June 12, 2025.
- We focused on the geographic distribution of patient reported and enrolled in these trials vis-à-vis and the number of site in the trial across these regions.
- FDA approvals were identified using the publicly available Hematology/Oncology Approvals and Safety Notifications archive. Data collection cutoff was June 2025.
- Data for each pivotal trial was extracted from PubMed (published primary reports of these trials were used) and ClinicalTrials.gov, including key variables such as drug name, approval type, phase, sponsor, trial design, duration, number of sites, participating countries, and total enrollment. Trials without full publications or clinical registry entries were excluded.
- Global enrollment was analyzed by country income classification as per World Bank classification and region to assess trial diversity and representation.

Conclusion

- 51% of clinical trials did not report details on patient contribution from various countries/regions and less (36%) reported subgroup analysis for the primary endpoint pertaining to population.
- HICs comprised 70% of all sites that participated in the trial but enrolled only 59% of all patients (P < 0.0001). This contribution, particularly US/Canada appeared unfavorable.
- Globalization of clinical trials has altered global representation in registrational studies in oncology.
- Understanding these trends are key to devising strategies to ensure equity and generalizability of research

Results

147 Approvals
103 Therapeutic Agents
155 Trials Reported
69,271 Patients

Figure 1. Correlation matrix showcasing the strong association of trial size (patients), number of sites and number of countries involved in clinical trials.

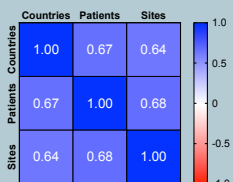


Table 1. Key characteristics of trials included in the analyses

Approvals	N	%	Approval Category	N	%
Agents Category			Approval Category		
Cytotoxics/ADCs	23	16	Full	113	77
Targeted Therapy	69	47	Accelerated	34	23
Immunotherapy	55	37	Study Phase		
Cancer Type/Staging			I, I/II, II	56	38
Hematological	37	25	II/III, III	91	62
Solid Tumors	110	75	Randomized Trial		
Locoregional	12	11	Yes	91	62
Metastatic	98	89	No	56	38

- 140 of 155 (90%) of trials were global in nature and run across a median of 16 countries (range: 2-38) involving a median of 114 sites (range: 1-597).
- 136 (97%), 133 (95%), 114 (81%), 83 (59%), 62 (44%) and 14 (%) trials were performed in North-America (US/Canada), Europe, Asia, Australia, South-America and Africa, respectively.
- Number of sites in participating HICs was significantly higher than others (median: 25 vs 16; P = 0.008). See Figure 2.

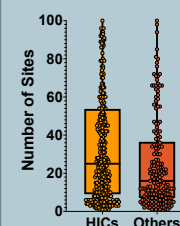


Figure 2. Comparison of number of sites.

- 72 of 140 (51%) global trials did not report on any geographic contribution.
- 51 of 140 (36%) global trials that reported on geographic distribution performed sub-group analysis of primary endpoint based on this.

- Of trial that reported on geographic distribution, HICs had significantly lower patient participation relative to number of sites compared to others:

- HICs constituted 69.7% of all sites but contributed 58.8% of all patients to these trials compared to others (30.3% sites with 41.2% patients) (P < 0.0001)
- In trials where US/Canada participated, and geographic distribution was reported:
- US/Canada constituted 24.3% of all sites but contributed 15.6% of all patients to these trials compared to others (75.7% sites with 84.4% patients) (P < 0.0001)

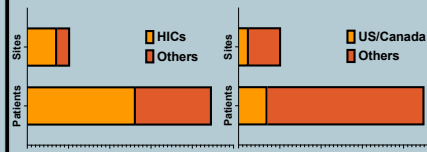


Figure 3. HICs (OR: 0.62) and US/Canada (OR: 0.57) enrolled relatively lower number of patients to trials compared to the sites involved in trials.